



Lab Worksheet 3

CSE461: Introduction to Robotics

Department of Computer Science and Engineering

Lab 03: Introduction to Raspberry PI microcontroller and control an LED with an ultrasonic sensor using the board.

I. Topic Overview

The lab is designed to introduce the basics of a microcontroller using Raspberry PI development board and how to use sensors using the board. The students will learn how to use the GPIO - General Purpose Input Output pins for interfacing simple I/O devices like LED and ultrasonic sensors with the board. We will also be discussing the process of measuring distance using an ultrasonic sensor with the help of a Raspberry Pi. The aim of this lab is to demonstrate how to use an ultrasonic sensor with Raspberry Pi to measure distance accurately.

II. Learning Outcome

After this lab, students will be able to:

1. Know the various pins of the Raspberry PI board and their functions.
2. Build a simple circuit with the board and I/O devices.
3. Code to configure the GPIO pins to control different circuits for different usages.
4. Understand the basic principles of ultrasonic sensors and how they can be used to measure distance.
5. Learn how to install Python libraries on the Raspberry Pi and how to create and run Python programs using the terminal.

III. Materials

- Raspberry PI development board
- LED (Light Emitting Diode)
- Ultrasonic Sensor(HC-SR04)
- Resistor (appropriate value for current limiting, typically around 220 ohms)
- Jumper wires

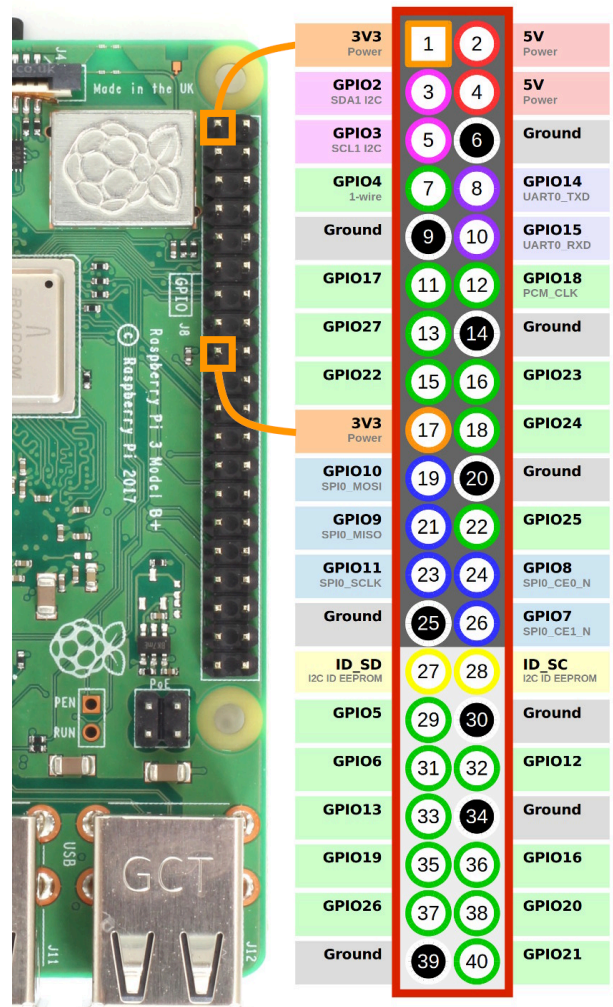
IV. Raspberry Pi Pins:

The Raspberry PI 4B has the following types of pins:

1. Power Pins:

- 5V and 3.3V:** Provide regulated voltage output.
- GND:** Ground pins.

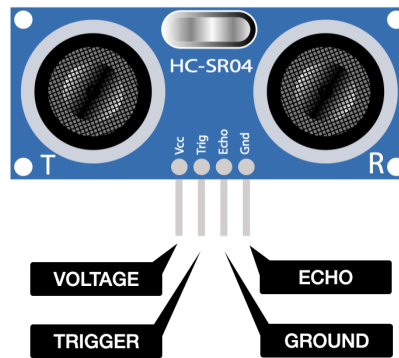
- GPIO Pins:** The general purpose input and output pins. Some of these GPIO pins can be used for following different communication protocols like UART, I2C and SPI to communicate with other devices. The GPIO pins work at a high voltage of 1.8V-3.3V maximum.



V. Experiment 2: Measuring distance using ultrasonic sensor.

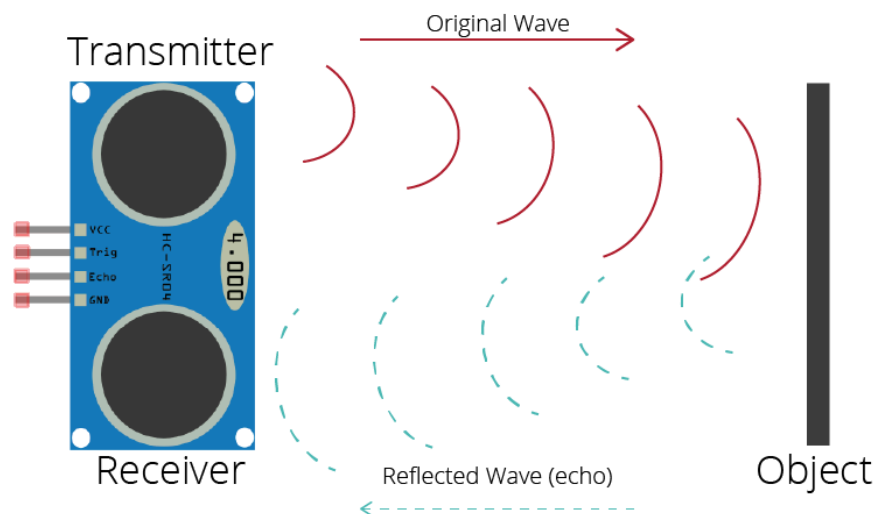
Description the components:

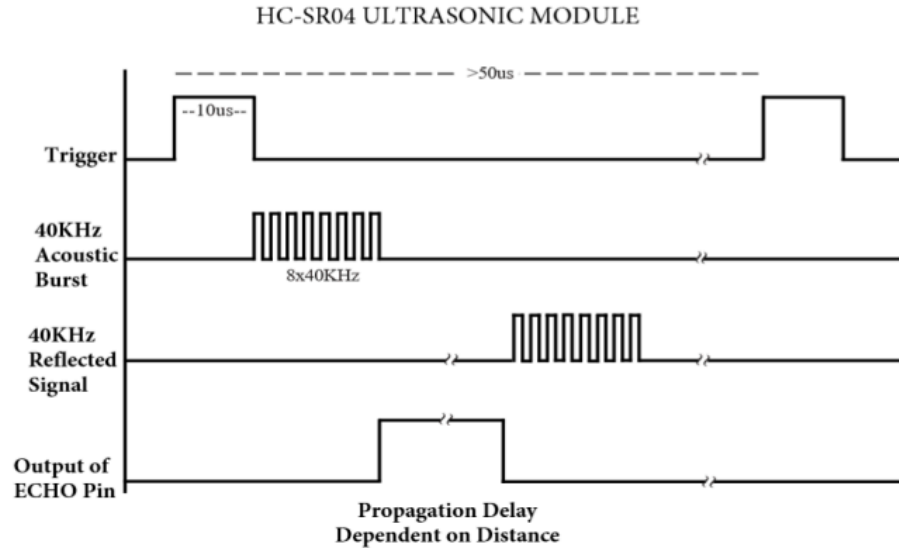
- a. **Ultrasonic Sensor:** Ultrasonic sensors are a common type of sensor that are used for measuring distances by sending out sound waves and measuring the time it takes for them to bounce back.



- **Pin1 (Vcc):** This pin provides a +5V power supply to the sensor.
- **Pin2 (Trigger):** This is an input pin, used to initialize measurement by transmitting ultrasonic waves by keeping this pin high for 10us.
- **Pin3 (Echo):** This is an output pin, which goes high for a specific time period and it will be equivalent to the duration of the time for the wave to return back to the sensor.
- **Pin4 (Ground):** This is a GND pin used to connect to the GND of the system.

Working mechanism of an ultrasonic sensor:





- A microcontroller is used for communication with an ultrasonic sensor.
- To begin measuring the distance, the microcontroller sends a trigger signal to the ultrasonic sensor. The duty cycle of this trigger signal is $10\mu s$ for the HC-SR04 ultrasonic sensor.
- When triggered, the ultrasonic sensor generates eight acoustic (ultrasonic) wave bursts and initiates a time counter.
- As soon as the reflected (echo) signal is received, the timer stops. The output of the ultrasonic sensor is a high pulse with the same duration as the time difference between transmitted ultrasonic bursts and the received echo signal.

1. Setting up the circuit:

- Connect the VCC pin of the ultrasonic sensor to the 5V pin of the board.
- Then the GND pin with the ground pin of the board.
- Connect the TRIG (trigger pin) and the ECHO pin with GPIO pin 21 and 20.

Note: The **ECHO pin outputs 5V** which will damage the GPIO pin of the Raspberry PI as it can only take 3.3V at maximum. So in order to convert it to a safer level we will use resistors using the voltage divider rule shown in the figure below. The 5V from the Echo pin will first need to be converted to 3.3V and then connect it to the GPIO pin. Then the voltage needs to be dropped to the ground. The ratio of the voltage division has to be 1:2 where we will be using one 220 ohm resistor connected to 5V and two 220 ohm resistors to GND.

Voltage Divider Formula:

The general formula for a voltage divider is:

$$V_{out} = V_{in} \times \frac{R_1}{R_1 + R_2}$$

where:

- $V_{in} = 5V$
- $V_{out} = 3.3V$
- R_1 is the resistor connected to 5V
- R_2 is the resistor connected to GND

Using three 220 ohm resistors:

One resistor (220 ohm) as R_1

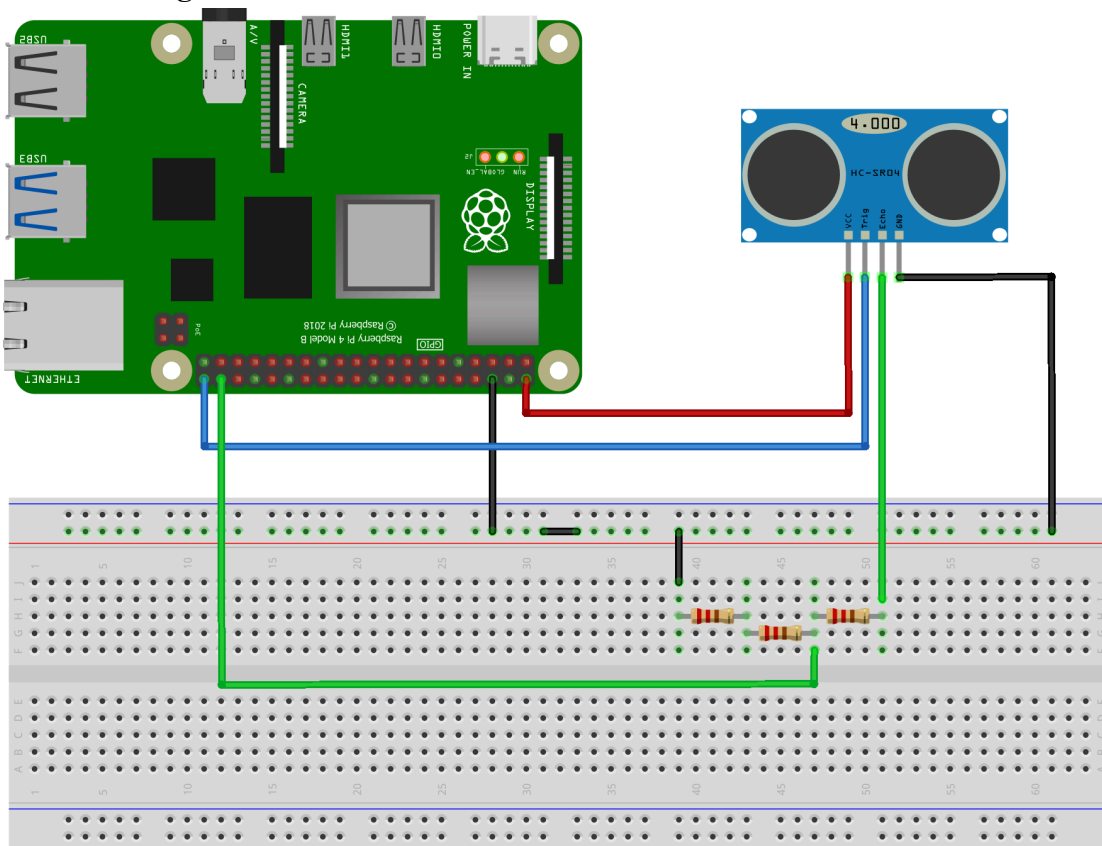
Two resistors (220 ohm each) in series as $R_4 = R_2 + R_3 = 220 \text{ ohm} + 220 \text{ ohm} = 440 \text{ ohm}$

Now, applying the voltage divider formula:

$$V_{out} = V_{in} \times \frac{R_1}{R_1 + R_4} = 5V \times \frac{220\Omega}{220\Omega + 440\Omega} \approx 3.33V$$

This is close to 3.3V, which is safe for the Raspberry Pi GPIO pins.

2. Circuit diagram:



*How this works: The ultrasonic sensor works at 40hz. The speed of sound at this frequency in air is 343ms. Since the scope of our sensor is low, we will measure the distance in cm. Hence we convert the speed to $(343*100)$ or 34300 cms.*

VI. Lab Task: Turn an LED on if the distance measured by the ultrasonic sensor is less than 5CM.

VII. References:

1. <https://datasheets.raspberrypi.com/rpi4/raspberry-pi-4-datasheet.pdf>